

J: Courage is Contagious

Source file name: `courage.c`, `courage.cpp`, `courage.java`, or `courage.py`

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On the plains of *Tiě Shā* --- the Land of Iron Sand --- the wind does not blow: it howls. The Northern Barbarians have crossed the Frozen River for the tenth time in a generation, and this time they are not coming to plunder. They are coming to stay. General **Wei Zhan** gathers his soldiers at the edge of the abyss: behind them, the Han Empire; ahead, a tide of steel and fury. There is no wall. There are no reinforcements. There are only men --- and the question each one asks in silence as the horizon darkens: *How many of my brothers are going to fight?*

General Wei Zhan knows his men well. Each soldier i carries a bravery threshold k_i : at the start of each round he looks at *how many soldiers are fighting in total*, and he will be fighting in the next round if and only if that total is at least k_i . It does not matter *who* is fighting, only *how many*. At dawn, some number of warriors draw their swords; from that moment the count updates round by round --- soldiers whose threshold is met join the fight, soldiers whose threshold is not met stand down --- until the number of fighters no longer changes.

Because every soldier reacts only to the *total* number of fighters, the outcome depends solely on that number at dawn, never on the particular soldiers who started. If c soldiers are fighting in a round, then in the next round the soldiers fighting are exactly those with $k_i \leq c$; the process repeats until the count is stable.

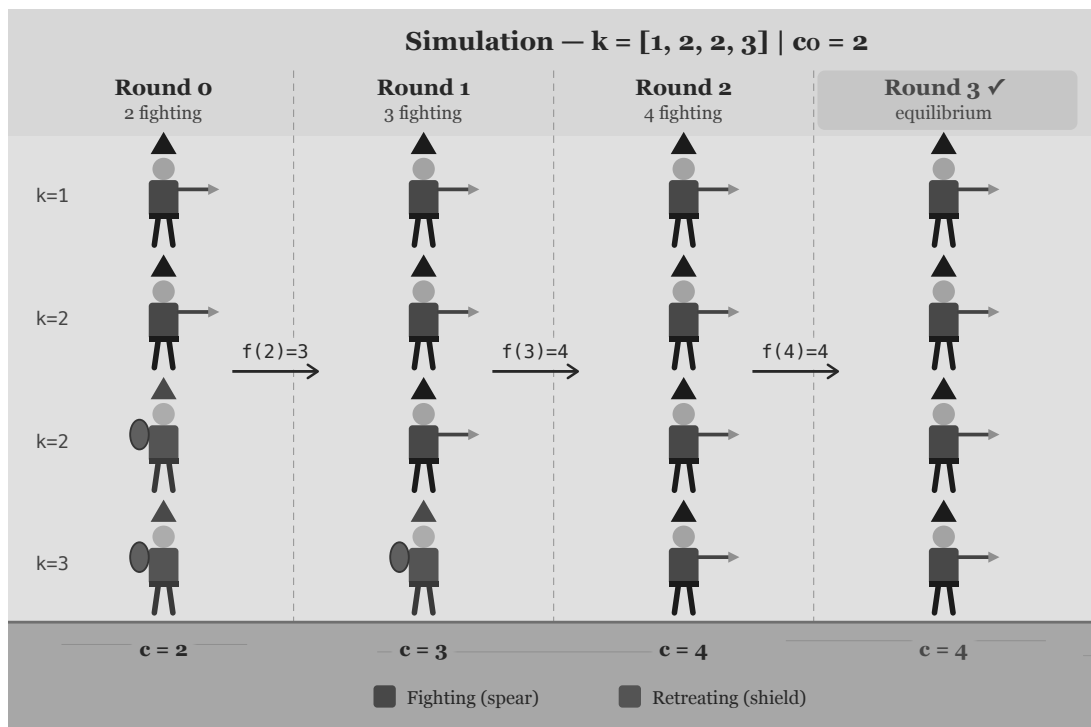


Figure 4: Simulation for $N = 4$ soldiers with thresholds $k = [1, 2, 2, 3]$ and initial count $c_0 = 2$. The number of fighters grows $2 \rightarrow 3 \rightarrow 4$ and then stabilizes: all 4 soldiers are fighting.

For example, consider $N = 4$ soldiers with thresholds $k = [1, 2, 2, 3]$ and an initial count of $c_0 = 2$ warriors at dawn. With 2 fighting, every soldier with threshold ≤ 2 fights next --- that is three soldiers (thresholds 1, 2, 2), so the count rises to 3. With 3 fighting, soldiers with threshold ≤ 3 fight --- all four --- so the count rises to 4.

With 4 fighting, again all soldiers with threshold ≤ 4 fight: still 4. The count is stable. The answer is **4**.

Before dawn, Wei Zhan must know the outcome of every possible initial formation. Given only the number c_0 of soldiers fighting at daybreak, how many will be fighting when the plains of Tiě Shā reach their equilibrium?

Input

The input consists of multiple test cases. Each test case begins with a line containing two integers N and Q ($1 \leq N \leq 10^5$, $1 \leq Q \leq 10^5$), the number of soldiers and the number of queries, respectively. The next line contains N integers $k_1, k_2, \dots, k_N$ ($0 \leq k_i \leq N$), the bravery threshold of each soldier. The following line contains Q integers c_0 ($0 \leq c_0 \leq N$), each representing an initial count of soldiers fighting at dawn. The sum of N over all test cases and the sum of Q over all test cases each do not exceed 10^6 . The input ends with a line containing 0.

The input must be read from standard input.

Output

For each query c_0 , print a single integer: the number of soldiers fighting when the battlefield reaches equilibrium. Print one result per line.

The output must be written to standard output.

Sample Input	Sample Output
4 2	4
1 2 2 3	0
2 0	5
5 3	0
1 2 2 3 4	1
2 0 1	
0 0	